

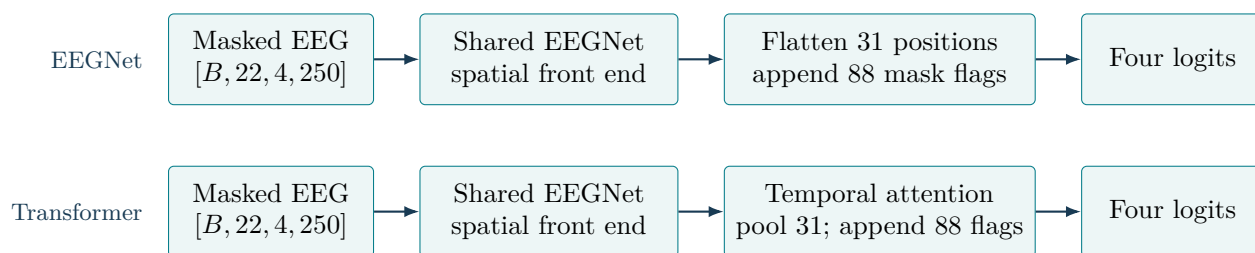
Spatial EEGNet and spatial Transformer

A teaching guide to the reproducible candidate cohort

Scope. Two end-to-end raw-EEG classifiers in the separate `encoder_candidates` study: `spatial_eegnet` and `spatial_transformer`. This guide follows the real, frozen reproducible-v1 cohort. Both systems classify four motor-imagery classes for one participant at a time. The name “spatial Transformer” refers to its EEGNet spatial front end; the Transformer attention itself is over a short temporal sequence.

1 The models at a glance

The question is whether replacing EEGNet’s flattened temporal readout with one small Transformer encoder layer changes clean validation performance. The front end first mixes electrode signals into learned features. The Transformer then relates 31 time positions in those features. Neither arm uses the legacy Tucker representation or attention over 22 electrode tokens.



Property	<code>spatial_eegnet</code>	<code>spatial_transformer</code>
Shared front end	EEGNet temporal, spatial, and separable convolutions	Same modules and identical initialization for a given seed
Readout sequence	$[B, 31, 16]$, flattened to 496 values	Project to $[B, 31, 32]$, one temporal Transformer layer, mean pool
Availability input	88 Boolean channel-window flags	Same 88 flags after temporal pooling
Trainable parameters	3,796	11,092
Role in comparison	Reference arm	Complete alternate head, not an isolated attention ablation

Read the word “spatial” carefully. EEGNet’s depthwise convolution spans the 22-electrode axis. Once that operation runs, its 16 feature maps are learned mixtures, not individually identifiable electrodes. The Transformer applies self-attention over the 31 pooled temporal positions.

Reading map. Sections 2–4 trace the inputs and both architectures; section 5 explains loss and evaluation; sections 6–7 cover results, prior models, and novelty; section 8 records provenance and source locations.

2 Dataset, trial construction, and normalization

One example

The cohort uses BCI Competition IV dataset 2a: nine participants, four classes (left hand, right hand, feet, tongue), 22 EEG electrodes, sampled at 250 Hz. The default files are the participants' T-session recordings. A trial starts at the class cue and contains 1,000 samples, or four seconds. Artifact marked, out-of-bounds, or nonfinite trials are excluded by the loader; counts are recorded in provenance. The canonical signal before windowing is $[N, 22, 1000]$, with integer label $y \in \{0, 1, 2, 3\}$.

Stage	Operation and resulting shape
Read	Accepted cue-aligned EEG: $[N, 22, 1000]$
Filter	2–30 Hz Butterworth SOS, order argument 4, forward–backward; each electrode and one-second window filtered separately
Window	Split the 1,000 samples into four adjacent windows of 250 samples: $[N, 22, 4, 250]$
Normalize	Training-trial/channel/time mean and population standard deviation; standard deviation floor 10^{-8}
Input	Float tensor $X \in \mathbb{R}^{B \times 22 \times 4 \times 250}$
Mask	Boolean availability $M \in \{0, 1\}^{B \times 22 \times 4}$; 1 means present in that whole one-second window
Output	Four unnormalized class scores (logits), shape $[B, 4]$

For each channel c , the training partition alone determines

$$\mu_c = \frac{1}{|\mathcal{T}|1000} \sum_{i \in \mathcal{T}, t} R_{ict}, \quad \sigma_c = \max \left(\sqrt{\frac{1}{|\mathcal{T}|1000} \sum_{i \in \mathcal{T}, t} (R_{ict} - \mu_c)^2}, 10^{-8} \right), \quad X_{ict} = \frac{R_{ict} - \mu_c}{\sigma_c}.$$

The saved training statistics are applied unchanged to validation. Samples are split within each participant and T session, approximately 60% train, 20% validation, and 20% held out, stratified by class for each seed. There are nine participants and three seeds, or 27 paired participant/seed tasks.

Removing unavailable values safely

Before any convolution or channel mixing, the models select observed values:

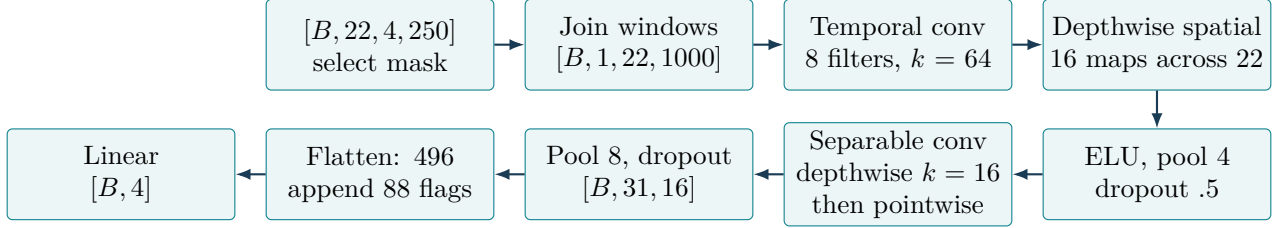
$$\tilde{X}_{bcpr} = \begin{cases} X_{bcpr}, & M_{bcp} = 1, \\ 0, & M_{bcp} = 0. \end{cases}$$

The implementation uses `torch.where`. This matters if a hidden entry is NaN: multiplying it by zero still leaves NaN. A zero after normalization is the training mean value in raw units. Each trial/window must retain at least one electrode.

Training is **full input only**: every one of the 88 flags is 1 during optimization and epoch selection. Masked validation conditions are diagnostics. Passing a mask to the model does not by itself train robustness to missing electrodes.

3 Model 1: the mask-conditioned spatial EEGNet

This arm wraps the project’s EEGNet classifier with fixed settings $F_1 = 8$, depth multiplier $D = 2$, and $F_2 = 16$. The implementation follows EEGNet’s compact temporal / depthwise spatial / separable convolution pattern [2]. It learns all filters and the classifier together by class cross-entropy.



Shapes and purpose

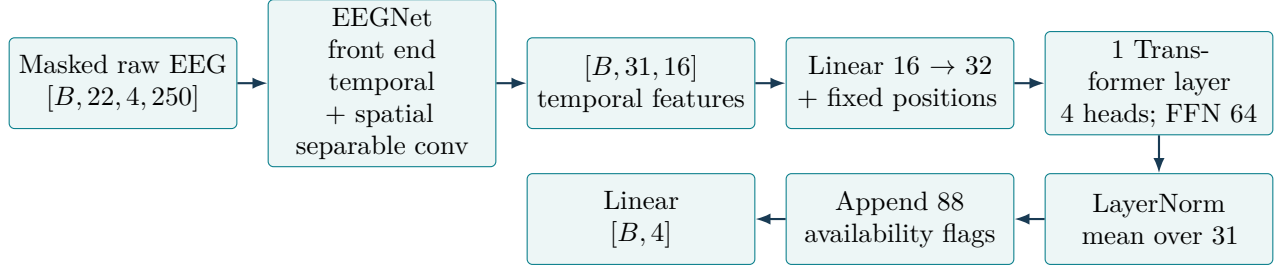
Operation	Output	Interpretation
Select mask, reshape	$B \times 1 \times 22 \times 1000$	Four adjacent windows are rejoined before learned convolution
Temporal conv + BatchNorm	$B \times 8 \times 22 \times 1000$	Length-64 filters run at every electrode
Depthwise spatial conv + BN	$B \times 16 \times 1 \times 1000$	Two electrode-spanning filters per temporal map
ELU, average pool 4, dropout	$B \times 16 \times 1 \times 250$	Nonlinearity and first time reduction
Depthwise temporal conv $k=16$, pointwise mix, BN	$B \times 16 \times 1 \times 250$	Temporal refinement, then mix feature maps
ELU, average pool 8, dropout	$B \times 16 \times 1 \times 31$	Ordered temporal summary positions
Squeeze and transpose	$B \times 31 \times 16$	Time-major sequence exposed by <code>forward_features</code>
Flatten + 88 flags	$B \times 584$	$31 \cdot 16 + 22 \cdot 4$ classifier inputs
Linear	$B \times 4$	Four class logits

The convolutions use padding that preserves time length before pooling. The temporal filters run over the concatenated 1,000-sample sequence with kernel width 64. Filtering is window-local, but the learned convolution can mix information across the joins between windows. The spatial convolution collapses the electrode axis; it does not produce one output token per electrode. Dropout rate is 0.5. After each optimizer step, the EEGNet arm clips spatial-filter max norm to 1.0 and final-classifier row max norm to 0.25.

The mask is flattened in canonical electrode-then-window order and appended after the 496 signal values. Since training masks are all ones, this head learns with a constant availability vector during fitting. At masked evaluation, the vector informs the classifier which inputs were zeroed, while the front-end filters still see those zeros.

4 Model 2: EEGNet front end plus a compact temporal Transformer

The arm constructs the exact EEGNet initialization for a given seed and keeps its convolutional feature layers. The EEGNet arm and Transformer arm therefore start with identical front-end tensors. They then optimize independently; the shared start is not a frozen feature extractor.



Temporal positions and self-attention

Each of the 31 front-end positions has 16 values. A learned linear map gives 32 values, then a fixed sinusoidal position code is added. This tells the attention layer which temporal position a vector came from. One batch-first, pre-layer-normalized encoder layer uses four attention heads, a 64-wide GELU feed-forward sublayer, and dropout 0.1. A final LayerNorm is applied before averaging positions. The result is 32 values; appending the same 88 flags gives 120 inputs to the class layer.

For one attention head, the scaled dot-product operation is

$$\text{Attn}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V, \quad Q = XW_Q, K = XW_K, V = XW_V.$$

Here rows of X are time positions (31 rows), not electrodes. Multi-head attention computes this with four learned projections, combines the head outputs, and uses residual connections and layer normalization. The PyTorch encoder layer also contains the position-wise feed-forward block

$$\text{FFN}(x) = W_2 \text{GELU}(W_1 x + b_1) + b_2, \quad 32 \rightarrow 64 \rightarrow 32.$$

With pre-normalization, each sublayer adds its input through a residual path. The Transformer output sequence remains $[B, 31, 32]$ until mean pooling. Its training step clips shared spatial filters to max norm 1.0 but does not clip the final classifier; this is one regularization difference from EEGNet.

Setting	Declared value
Projection width / position code	16 → 32 / fixed sinusoidal for positions 0–30
Encoder	One layer; four heads; batch-first; norm-first
Feed-forward / activation	Width 64 / GELU
Dropout / final readout	0.1 / LayerNorm, mean of positions
Classifier inputs	32 pooled values plus 88 Boolean mask flags
Parameters	11,092 trainable; positional code is a fixed buffer

This is a complete head comparison: projection, temporal attention, positional code, pooling, and classifier change relative to EEGNet. It is not a controlled test of attention alone or a test of whether temporal order helps; a parameter- matched head and mean/linear alternatives would be needed for that claim.

5 Training objective, epoch selection, and evaluation

The class scores $z_i \in \mathbb{R}^4$ become probabilities by softmax:

$$p_{ik} = \frac{\exp(z_{ik})}{\sum_{j=0}^3 \exp(z_{ij})}.$$

For training batch $\mathcal{B}$, the unweighted cross-entropy is

$$\mathcal{L}_{\text{CE}}(\theta) = -\frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \log p_{i, y_i}, \quad z_i = f_{\theta}(\tilde{X}_i, M_i).$$

The code passes logits to PyTorch’s numerically stable cross-entropy. There is no reconstruction, tensor, or contrastive term. The gradient on each logit is $(p_{ik} - \mathbf{1}[k = y_i])/|\mathcal{B}|$: it raises the true-class score relative to the alternatives. Both arms train end-to-end.

Training setting	Value
Optimizer	AdamW, minibatch 32, weight decay 10^{-4}
Learning rate	10^{-3} maximum; linear warmup for 10 epochs, cosine decay to zero
Training budget	At most 250 epochs; minimum 75 before early stopping
Stopping	50 epochs without validation improvement
Gradient clipping	Global norm at most 1
Model selection	Full-input validation balanced accuracy; log-loss tie-break, then earlier epoch
RNG	Fixed per-task/per-fit seeds and epoch-specific data order; deterministic algorithms

At each epoch, the best checkpoint key is (validation balanced accuracy, negative validation log-loss). Balanced accuracy averages class recalls:

$$\text{BA} = \frac{1}{4} \sum_{k=0}^3 \frac{\#\{i : y_i = k, \hat{y}_i = k\}}{\#\{i : y_i = k\}}.$$

The test partition is not used for epoch selection and no test scores are included in this candidate study’s outputs or findings.

Paired masked validation

Each selected full-input checkpoint is also scored on the same five validation conditions: full 22, random static 16, random dynamic 16, random static 6, and random dynamic 6. Non-full conditions use five paired mask repeats. Static masks retain one subset for all windows; dynamic masks can change between windows. All arms share the same validation trial IDs and mask hashes. These are diagnostic evaluations after full-only training; they do not establish robust training for sensor loss.

Model selection and all cohort comparisons use validation. Since validation also selects each checkpoint, the paired bootstrap intervals below are exploratory, not independent confirmation. No comparison to a published score is valid unless data split, preprocessing, selection, and scoring are matched.

6 What the completed cohort found

The frozen cohort has 27 real participant/seed tasks (A01–A09, seeds 0–2). The evidence review verified all task and fit identities, 675 declared artifact hashes, paired-arm records, and selected-checkpoint replay. It reports validation metrics only. For each arm, seed scores are averaged within participant, then participants are equally weighted.

Arm	Clean val. BA	Paired gain vs EEGNet	Positive participants	Clean val. log-loss
Spatial EEGNet	59.50%	Reference +5.37 pp	—	1.057
Spatial Transformer	64.87%	[−0.42, +12.00]	7/9	0.993

The bracket is the exploratory paired 95% participant-bootstrap interval (2,000 resamples). The Transformer satisfies the frozen practical screen (mean gain at least 2 percentage points and at least six positive participants), but its interval includes zero. Its average log-loss improvement is 0.065 (reference minus candidate), also with an interval crossing zero ([−0.099, +0.257]). It has 2.9 times as many trainable parameters as the EEGNet arm. The Transformer therefore remains inconclusive.

For the four missing-electrode conditions, Transformer balanced accuracy is essentially tied with EEGNet at 16 electrodes (paired changes −0.02 pp static and +0.22 pp dynamic) and trails by 0.62 pp in both six-electrode conditions. Its log-loss is worse than EEGNet in all four conditions, by 0.567–0.757. Neither result supports a missing-electrode robustness claim.

The mean clean train BA is 86.84% for EEGNet and 76.38% for the Transformer; their validation BA is 59.50% and 64.87%, respectively. The train-minus-validation BA gaps are 27.34 pp for EEGNet and 11.51 pp for the Transformer. For the Transformer, validation log-loss is 0.436 higher than train. These differences describe the selected checkpoints; they do not identify a cause. The Transformer has a smaller BA gap here, but its higher validation BA remains inconclusive and does not establish stronger generalization.

These figures are **not test accuracy, published benchmark scores, or state-of-the-art comparisons**. They are validation outcomes under one within-participant T-session split protocol, after validation-based epoch selection. The available evidence supports a tentative full-input signal for the Transformer and no improvement in masked log-loss.

The study identity is b72da29fe20c355bed86810c5a865595678e2b476f0f91f7083437da163246dc; the configuration SHA-256 is:

b0ede0abf81204bcc447b00f2fba62ae87ca7a55fbe4a798ceb4740cbda62ea6

The research implementation was untracked during the experiment, so this identity and the source-file digest manifest in the report are more accurate provenance than assigning it a Git commit.

7 How these models relate to published work

EEGNet introduced a compact EEG-specific use of temporal convolution followed by depthwise spatial filtering and separable convolution [2]. The reference in this cohort is an EEGNet-style adaptation: it receives a four-window tensor, rejoins those windows before convolution, appends explicit availability flags, and uses this project’s participant-wise trial split and selection protocol. It is not a full reproduction of a published benchmark.

The Transformer head uses the standard encoder ideas of projected queries, keys and values, residual paths, position-wise feed-forward layers, and positional information [3]. EEG-specific models have also combined convolutional feature extraction with Transformer attention; EEG Conformer is one published example [4]. ATCNet combines convolution, temporal attention, and temporal convolution for motor imagery [5]. Therefore the broad idea “EEGNet-like convolution followed by Transformer attention” is not a new architecture claim for this code. The particular compact settings and comparison in this repository are an engineering adaptation and candidate evaluation, not a demonstrated methodological novelty.

Can we call this state of the art?

No. The cohort does not compare against published leaderboard values under a matched protocol, and it does not use a subject-disjoint split, cross-session test, or an untouched confirmatory cohort. BCI Competition IV results depend on session, trial window, preprocessing, data partition and metric; published scores cannot be ranked against the validation values above as if they shared one test set. The standard EEGNet paper, EEG Conformer, and ATCNet report distinct protocols and are useful architecture references, not directly comparable baselines for these results.

Architecture comparison

Model	Main architecture pattern	Relation to this study
Our spatial EEGNet	EEGNet-style temporal convolution, full-electrode depthwise spatial filtering, separable temporal convolution	Adds four-window input handling, zeroed unavailable samples, 88 mask flags, and this study’s participant split to a known EEGNet block.
Published EEGNet [2]	Compact temporal, depthwise spatial, and separable convolutions for EEG decoding	Architecture lineage for the reference; our preprocessing and validation protocol do not reproduce its paper benchmark.
Our spatial Transformer	EEGNet stem; one Transformer layer over 31 pooled positions; fixed sinusoidal positions; mean pool	Reuses the local EEGNet front end. Its short temporal readout and mask flags differ from a paper reproduction.
EEG Conformer [4]	Convolutional EEG stem followed by self-attention over local temporal features	Established convolution-plus-attention precedent; it has its own convolutional stem and readout. No scores are directly compared here.
ATCNet [5]	Convolution, temporal attention, and temporal convolution for motor imagery	A representative motor-imagery architecture family; its temporal-convolution head differs from this study’s one-layer Transformer head.

These are representative architecture families, not a leaderboard or a claim that this short bibliography covers every relevant model. Their papers are useful lineage and design references; their numeric scores are intentionally omitted because train/test protocols differ.

What is and is not new here

The guide documents two fixed arms within this repository’s candidate study. The spatial EEGNet is an EEGNet architecture adaptation for raw masked four-window inputs. The spatial Transformer is that same spatial convolutional front end with a small temporal Transformer readout. It does not introduce a new attention mechanism, new tensor representation, or established new architecture family. A complete literature search was not conducted, so this is a cautious scope statement rather than an exhaustive priority claim.

Claim	Supported by this study?	Reason
Transformer improves clean validation BA in this cohort	Tentative only	+5.37 pp mean; interval crosses zero and validation chose epochs
Transformer improves electrode-loss robustness	No	Masked log-loss is consistently worse; models train full-only
New EEG architecture or SOTA	No	Adapted known components; no matched published benchmark or test comparison
Generalizes to unseen people/sessions	No	Split is within each person’s T session

8 Reproducibility and code map

The immutable study used the configuration SHA and study ID listed on page 6. Its report preserves per-file SHA-256 values, package versions, fit records, checkpoints, histories, predictions, and task identities. The reviewed cohort was CPU-only; it makes no claim about CUDA repeatability. Source changes require a new study identity and output directory.

Source	Responsibility
<code>configs/</code> <code>encoder-candidates-reproducible.json</code>	Frozen real cohort configuration
<code>inm/encoder_candidates/data.py</code>	Read-only split verification, filtering inputs, training-only normalization
<code>inm/baselines/eegnet.py</code>	Shared EEGNet convolutional model and shape path
<code>inm/encoder_candidates/spatial.py</code>	SpatialEEGNet reference and the separate filter-bank arm
<code>inm/encoder_candidates/transformer.py</code>	Temporal Transformer over EEGNet sequence
<code>inm/encoder_candidates/training.py</code>	Cross-entropy, deterministic fitting, validation selection, checkpoint replay
<code>inm/encoder_candidates/reporting.py</code>	Paired evidence checks and validation aggregation
<code>docs/candidate-followup/findings.md</code>	Recomputed cohort outcomes and interpretation limits
<code>results/</code> <code>encoder-candidates-reproducible-v1/</code> <code>report/evidence.json</code>	Frozen study identity, source hashes, package versions, task coverage

The main numerical limits are the small cohort (nine participants), reuse of validation for model selection, multiple contrasts in the broader five-arm study, and absence of a test score in the report. The paired intervals resample participants, not trials, but remain exploratory. The data and procedures do not support an online causal-filtering claim: the preprocessing is forward-backward and therefore offline.

References

- [1] C. Brunner et al. *BCI Competition 2008, Graz data set A* (BCI Competition IV, dataset 2a). Official dataset description. https://www.bbc.de/competition/iv/desc_2a.pdf.
- [2] V. J. Lawhern et al. *EEGNet: A Compact Convolutional Neural Network for EEG-based Brain-Computer Interfaces*. *Journal of Neural Engineering*, 15(5), 056013, 2018. DOI: 10.1088/1741-2552/aace8c. <https://arxiv.org/abs/1611.08024>; author implementation: <https://github.com/vlawhern/arl-eegmodels>.
- [3] A. Vaswani et al. *Attention Is All You Need*. *Advances in Neural Information Processing Systems*, 2017. <https://arxiv.org/abs/1706.03762>.
- [4] Y. Song et al. *EEG Conformer: Convolutional Transformer for EEG Decoding and Visualization*. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 2023. <https://doi.org/10.1109/TNSRE.2022.3230250>.
- [5] H. Altaheri, G. Muhammad, and M. Alsulaiman. *Physics-Informed Attention Temporal Convolutional Network for EEG-Based Motor Imagery Classification*. *IEEE Transactions on Industrial Informatics*, 19(2), 2249–2258, 2023. DOI: 10.1109/TII.2022.3197419. <https://doi.org/10.1109/TII.2022.3197419>.